

Tushar Bag

Assistant Professor

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Amaravati, Andhra Pradesh, India

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ResearchGate || Google Scholar

Research Interests

Algebraic Coding Theory · Quantum Error Correcting Codes

Academic Positions & Education

Assistant Professor (Grade I) <i>SRM University AP, Department of Mathematics, Amaravati, India</i>	<i>Sep. 2025 – Present</i>
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Postdoctoral Fellow <i>École Normale Supérieure de Lyon, Lyon, France</i>	<i>Jul. 2023 – Jun. 2025</i>
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Postdoctoral Fellow <i>Carleton University, Ottawa, Canada</i>	<i>Apr. 2022 – Apr. 2023</i>
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Postdoctoral Fellow <i>United Arab Emirates University, Abu Dhabi, UAE</i>	<i>Sep. 2021 – Apr. 2022</i>
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Doctor of Philosophy (Mathematics) <i>Indian Institute of Technology Patna, Patna, India</i> <i>CGPA: 8.71/10.</i>	<i>Aug. 2016 – Mar. 2020</i>
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Master of Science (Mathematics) <i>Indian Institute of Technology Kanpur, Uttar Pradesh, India</i> <i>CGPA: 6.6/10.</i>	<i>Jun. 2012 – May 2014</i>
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Bachelor of Science (Mathematics) <i>Ramakrishna Mission Vidyamandira, Calcutta University, India</i> <i>Percentage: 77/100.</i>	<i>Jun. 2009 – Apr. 2012</i>
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Class XII (Science) <i>Dhapdhara High School, WBBSE, India</i> <i>Percentage: 76/100.</i>	<i>Jun. 2007 – Apr. 2009</i>
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Under Review

2. Some New Non-binary Quantum Codes from One-generator Quasi-cyclic Codes (Revision Submitted)
1. Quantum Codes from Group Ring Codes (Revision Submitted)

Accepted

25. T. Bag and D. Panario, “Quasi-polycyclic and skew quasi-polycyclic codes over $\mathbb{F}_q$ ”, *Finite Fields and Their Applications* **101**, 102536, 2025.
24. K. Abdukhalikov, T. Bag and D. Panario, “One-generator quasi-cyclic codes and their dual codes”, *Discrete Mathematics* **346**(6), 113369, 2023.
23. H. Q. Dinh, B. P. Yadav, T. Bag, D. Panario and A. K. Upadhyay, “Self-dual and LCD double circulant and double negacirculant codes over a family of finite rings $\mathbb{F}_q[v_1, v_2, \dots, v_t]$ ”, *Cryptography and Communications*, 15(3), 529–551, 2022.
22. H. Q. Dinh, T. Bag, K. Abdukhalikov, S. Pathak, A. K. Upadhyay, R. Bandi and W. Chinnakum, “On a class of skew constacyclic codes over mixed alphabets and applications in constructing optimal and quantum codes”, *Cryptography and Communications*, 15, 171–198, 2023.
21. T. Bag, H. Q. Dinh, K. Abdukhalikov, A. K. Upadhyay, W. Yamaka, “Constacyclic codes over $\mathbb{F}_{q^2}[u]/\langle u^2 - w^2 \rangle$ and their application in quantum codes construction”, *Journal of Applied Mathematics and Computing*, 68, 3821–3834, 2022.
20. H. Q. Dinh, S. Pathak, T. Bag, A. K. Upadhyay, W. Yamaka, “Constacyclic codes over mixed alphabets and their applications in constructing new quantum codes”, *Quantum Information Processing* **20**(150), 2021.
19. H. Q. Dinh, T. Bag, P. K. Kewat, S. Pathak, A. K. Upadhyay, W. Chinnakum, “Constacyclic codes of length (p^r, p^s) over mixed alphabets”, *Journal of Applied Mathematics and Computing* **67**, 807–832, 2021.
18. H. Q. Dinh, T. Bag, A. K. Upadhyay, R. Bandi, R. Tansuchat, “A class of skew cyclic codes and application in quantum codes construction”, *Discrete Mathematics* **344**(2), 1–13, 112189, 2021.
17. H. Q. Dinh, S. Pathak, T. Bag, A. K. Upadhyay, R. Bandi, W. Yamaka, “On $\mathbb{F}_2RS$ -cyclic codes and their applications in constructing optimal codes”, *Discrete Mathematics* **344**(5), 2021.
16. H. Q. Dinh, T. Bag, S. Pathak, A. K. Upadhyay, W. Chinnakum, “Quantum codes obtained from constacyclic codes over a family of finite rings $\mathbb{F}_p[u_1, u_2, \dots, u_s]$ ”, *IEEE Access* **8**, 194082–194091, 2021.
15. H. Q. Dinh, S. Pathak, T. Bag, A. K. Upadhyay, W. Chinnakum, “A study of $\mathbb{F}_qR$ -cyclic codes and their applications in constructing quantum codes”, *IEEE Access* **8**, 190049–190063, 2021.
14. Y. Cao, Y. Cao, H. Q. Dinh, T. Bag, Y. Yamaka, “Explicit representation and enumeration of repeated-root $(\delta + \alpha u^2)$ -constacyclic codes over $\mathbb{F}_{2^m}[u]/\langle u^{2\lambda} \rangle$ ”, *IEEE Access* **8**(1), 55550–55562, 2020.

13. H. Q. Dinh, T. Bag, A. K. Upadhyay, R. Bandi, W. Chinnakum, “On the structure of cyclic codes over the ring $\mathbb{F}_q RS$ and applications in quantum and LCD codes constructions”, *IEEE Access* **8**(1), 18902–18914, 2020.
12. T. Bag, H. Q. Dinh, A. K. Upadhyay, W. Yamaka, “New non-binary quantum codes from cyclic codes over product rings”, *IEEE Communications Letters* **24**(3), 486–490, 2020.
11. T. Bag, H. Q. Dinh, A. K. Upadhyay, R. Bandi, W. Yamaka, “Quantum codes from skew constacyclic codes over the ring $\mathbb{F}_q[u, v]/\langle u^2 - 1, v^2 - 1, uv - vu \rangle$ ”, *Discrete Mathematics* **343**, 2020.
10. H. Q. Dinh, T. Bag, A. K. Upadhyay, M. Ashraf, G. Mohammad and W. Chinnakum, “Quantum codes from a class of constacyclic codes over finite commutative rings”, *Journal of Algebra and Its Applications* **19**(12), 2150003, 2019.
9. T. Bag, M. Ashraf, G. Mohammad, A. K. Upadhyay, “Quantum codes from $(1 - 2u_1 - 2u_2 - \dots - 2u_m)$ -skew constacyclic codes over the ring $\mathbb{F}_q + u_1\mathbb{F}_q + \dots + u_m\mathbb{F}_q$ ”, *Quantum Information Processing* **18**(9), 270, 2019.
8. T. Bag, A. Dertli, Y. Cengellenmis, A. K. Upadhyay, “Application of constacyclic codes over the semi local ring $\mathbb{F}_{p^m} + v\mathbb{F}_{p^m}$ ”, *Indian Journal of Pure and Applied Mathematics* **51**(1), 265–275, 2020.
7. T. Bag, A. K. Upadhyay, “Study on negacyclic codes over the ring $\mathbb{Z}_p[u]/\langle u^{k+1} - u \rangle$ ”, *Journal of Applied Mathematics and Computing* **59**(1–2), 693–700, 2019.
6. H. Islam, T. Bag, O. Prakash, “A class of constacyclic codes over $\mathbb{Z}_4[u]/\langle u^k \rangle$ ”, *Journal of Applied Mathematics and Computing* **60**(1–2), 237–251, 2019.
5. T. Bag, H. Islam, O. Prakash, A. K. Upadhyay, “A note on constacyclic and skew constacyclic codes over the ring $\mathbb{Z}_p[u, v]/\langle u^2 - u, v^2 - v, uv - vu \rangle$ ”, *Journal of Algebra Combinatorics Discrete Structures and Applications* **6**(3), 163–172, 2019.
4. T. Bag, S. Pathak, A. K. Upadhyay, “Classes of constacyclic codes of length p^s over the ring $\mathbb{F}_{p^m} + u\mathbb{F}_{p^m} + v\mathbb{F}_{p^m} + uv\mathbb{F}_{p^m}$ ”, *Beiträge zur Algebra und Geometrie* **60**(4), 693–707, 2019.
3. T. Bag, H. Islam, O. Prakash, A. K. Upadhyay, “A study of constacyclic codes over the ring $\mathbb{Z}_4[u]/\langle u^2 - 3 \rangle$ ”, *Discrete Mathematics, Algorithms and Applications* **10**(4), 1850056, 2018.
2. T. Bag, A. K. Upadhyay, “Skew cyclic and skew constacyclic codes over the ring $\mathbb{F}_p + u_1\mathbb{F}_p + \dots + u_m\mathbb{F}_p$ ”, *Asian-European Journal of Mathematics* **12**(5), 1950083, 2019.
1. T. Bag, A. K. Upadhyay, M. Ashraf, G. Mohammad, “Quantum codes from cyclic codes over the ring $\mathbb{F}_p[u]/\langle u^3 - u \rangle$ ”, *Asian-European Journal of Mathematics* **13**(1), 2050008, 2020.

Teaching Experience

Course	Title	Term / Role
FIC 117	Linear Algebra and Diff. Equa.	Even Sem. 2026, SRM AP
FIC 103	Calculus for Engineers	Odd Sem. 2025, SRM AP
MATH 1007	Elementary Calculus I	Fall 2022, Carleton University (Instructor)
MATH 1107	Linear Algebra I	Summer 2022, Carleton University (Instructor)
MA 201	Complex Analysis & PDE	Spring 2017, IIT Patna (TA)
MA 427	Abstract Algebra	Autumn 2016, IIT Patna (TA)

High School Teacher, Delhi Public School Bhagalpur, India

Aug. 2014 – Jan. 2016

Fellowships & Awards

- University Grants Commission (UGC) – Junior Research Fellowship, Government of India
- GATE – Junior Research Fellowship, Government of India
- Summer Research Fellowship Program, IIT Delhi, India
- Undergraduate Merit-cum-Means Scholarship, Government of India

Conferences & Workshops

- **ATMW Topology of Manifolds and Group Actions 2017**, University of Delhi, 9–22 Nov. 2017.
- **IST Topology 2017**, IIT Patna, 1–13 May 2017.
- **AIS h -Principle 2017**, Indian Statistical Institute, Kolkata, 22 May – 10 Jun. 2017.
- **NASI-TIMC Workshop on Coxeter Groups**, IIT Patna, 17–19 Dec.
- **Finite Fields and Galois Theory**, National Academy of Sciences, Allahabad, 22–23 Dec.

References

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